



香港大學
THE UNIVERSITY OF HONG KONG

Development and Integration of a Bionic Hand System Controlled by Muscle Deformation

Master of Science in Engineering in Innovative Design and Technology

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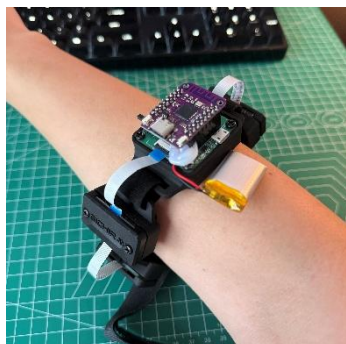
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Introduction

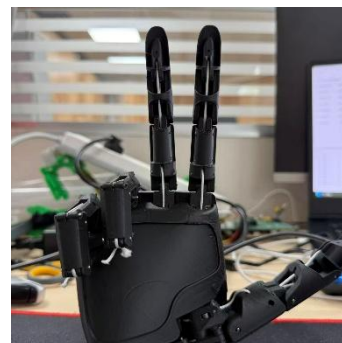
Data Acquisition



Raw signal data

Collect

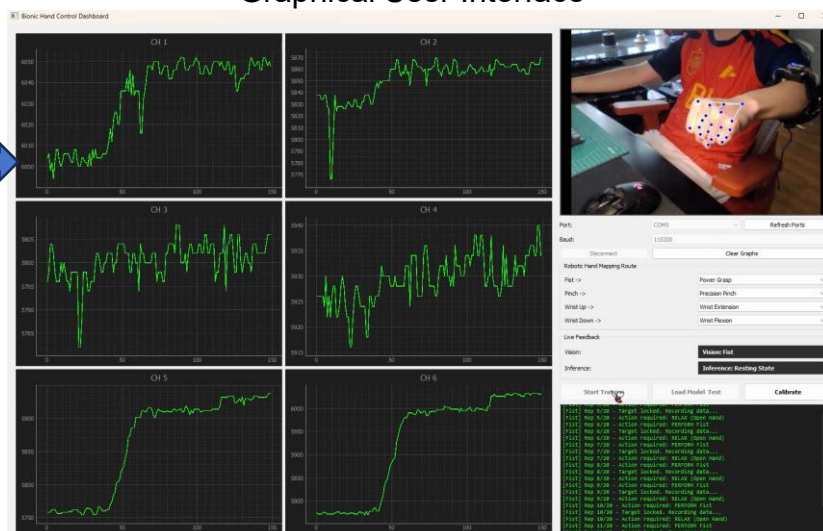
Execution



Control commands

Execute

Graphical User Interface



Algorithm models

Train & Load

System overview: An end-to-end data-driven bionic hand framework integrating:

- End-effector: 7-DOF open-source TetherIA tendon-driven hand [1]
- Physical sensing front-end: 6 channels Hall-effect muscle deformation armband
- Software: Multithreaded GUI with real-time ML/1D-CNN pipeline

Core Objective

To develop and validate a complete, low-cost bionic hand HMI system that bypasses the hardware and environmental limitations of surface electromyography (sEMG)

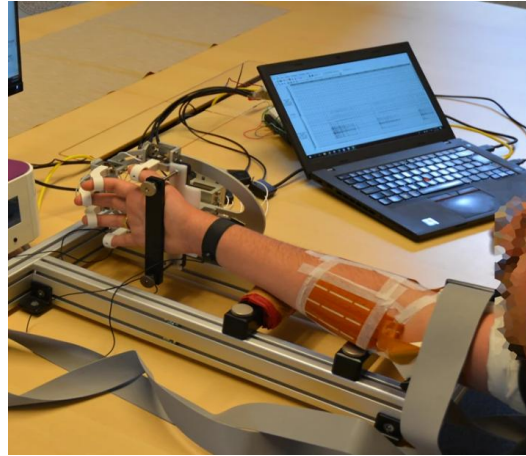
Background & Problem Statement



(3/i)



(3/ii)



(3/iii)



(3/iv)

Fig. (3/i) and (3/ii) demonstrated the bionic dexterous hand and sEMG armband from BrainCo [2], (3/iii) shows a high-density sEMG array system [3], (3/iv) Kinematics-Driven motion capture gloves [4]

Prohibitive Financial Barrier

- Extreme cost: Commercial hands (€8K-66K) and high-density/precision sEMG devices or motion capture gloves are too expensive for most of the customers, especially for amputee patients (not in gloves), as well as for research

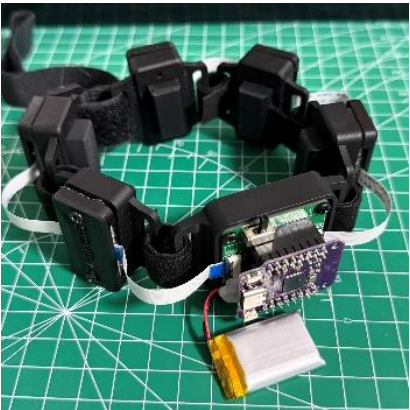
Physical Issues with sEMG sensors

- Signal instability: environmental elements (i.e., sweat, sensor shift, environmental noise) cause baseline drift during continuous use
- Information loss: Low fidelity ECG board (i.e., AD8232) filter out signals above 40Hz, losing vital muscle features

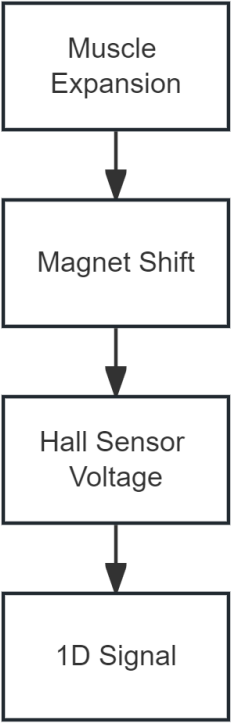
Limitations of Motion Capture Gloves

- Gloves (i.e., Manus) rely on direct joint retargeting rather than decoding underlying neuromuscular signals.
- Requires intact physical finger joints to measure movement, making them unusable amputee patients

Proposed Solutions: Muscle deformation sensing



(4/i)



(4/ii)

Metric/Feature	SOTA sEMG (e.g., Myo/HD-sEMG)	SOTA Muscle Deformation	Current system (6 channels Hall + TetherIA)
Sensing Mechanism	Microvolt bioelectric potentials (μV)	Physical muscle expansion (FMG/Magnetic)	Physical muscle expansion (6 channels Hall array)
Noise & Drift Immunity	Poor (i.e., Sensitive to impedance drift)	High (immune to moisture & bio-noise)	High (immune to moisture & bio-noise)
Signal processing overhead	High (Complex filtering, notch & PCA)	Moderate (Standard normalisation)	Low (Direct 1D signal)
Hardware Cost	High (€300-20,000+)	Moderate(€200-1,000)	Low (~¥500-€75)
Potential Application	Lab environments/Complex prostheses	Research & custom prototypes	Open-source research

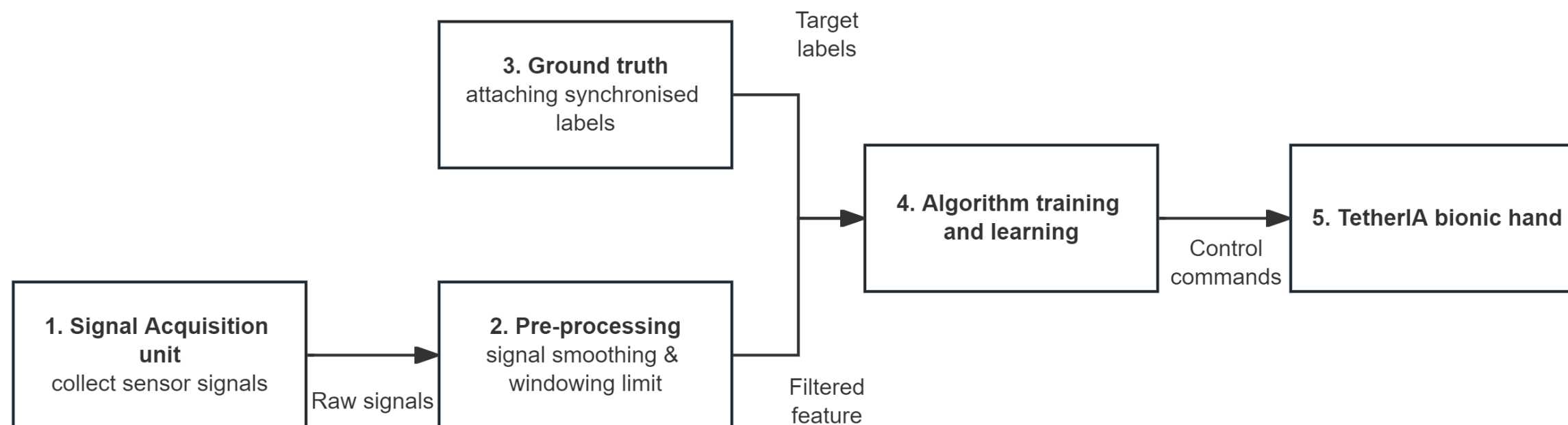
Mitigating Signal Drift: Replaces volatile microvolt signals with physical displacement. This bypasses sweat and impedance-induced noise, though large spatial shifts (e.g. armband rotation) remain a physical constraint

Automated Labelling: Incorporates a computer vision tracking pipeline strictly as an automated data labelling tool during collection, constructing a highly aligned dataset without human error

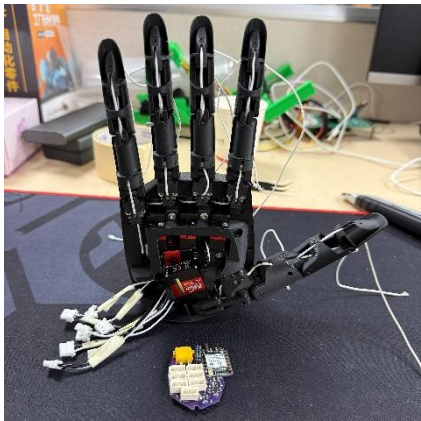
Fig. (4/ii) Sensing mechanism: Converting physical muscle expansion into 1D voltage signals. Fig. (4/i) The 6-channel Hall-effect armband



System Architecture



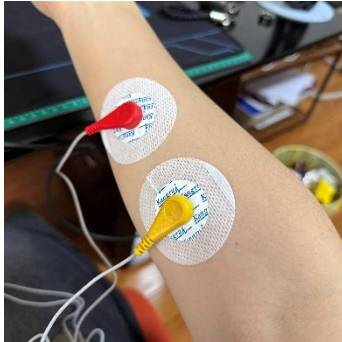
Hardware Assembly





Signal Acquisition Unit Iterations

Iteration 1



Low-fidelity ECG unit

Hardware: wet electrodes with hardware low-pass filters, breadboard-wired, USB connection

Failure mode: automatically truncated critical high-frequency signals

Iteration 2

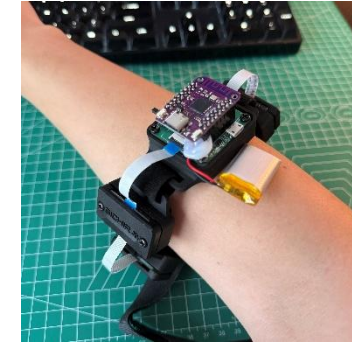


Dry electrode sEMG unit

Hardware: 2 channels dry electrode sEMG sensors, breadboard-wired, USB connection

Failure mode: likely to be influenced by sweat-induced baseline drift and environment noise

Iteration 3



Hall-effect muscle deformation sensing unit

Hardware: 6 channels Hall-effect sensors, custom PCB, wireless Bluetooth connection

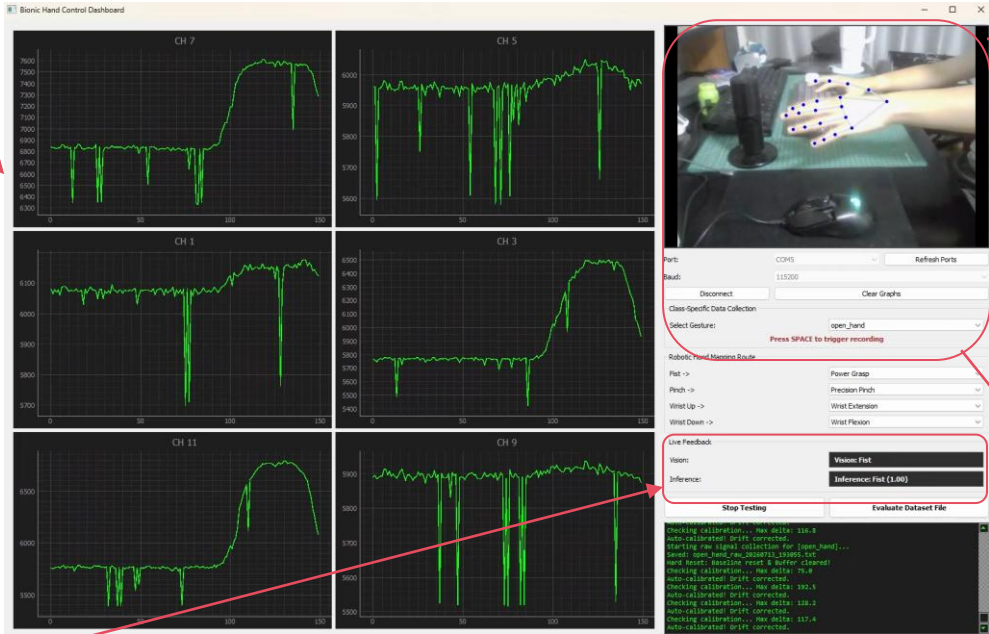
Engineering fix: bypassed sweat-induced drift and cable-related noise

Limitations: Thermal accumulation in chips during prolonged operations, cannot fully overcome spatial drift problem



Software Architecture & Data Collection Protocol

Real-time signals from signal acquisition unit



Live feedback from the video camera MediaPipe recognition and the model inference

GUI log, including connection state, data collection state etc.





Software Architecture & Data Collection Protocol

Old data collection



CSV file

- **Continuous recording in a single CSV file:** All gestures were collected consecutively in one long session
- **Fatigue-induced sensor drift:** Prolonged continuous muscle contraction led to severe physical baseline drift and data contamination
- **Lack of variance:** Uninterrupted repetitions failed to capture the necessary statistical diversity for robust model training

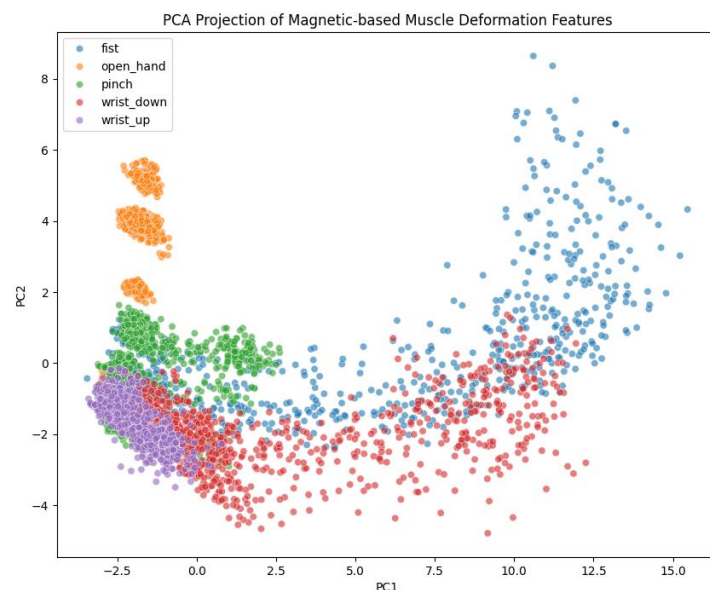
New data collection protocol

```
> archive_raw  
> fist  
> open_hand  
> pinch  
> wrist_down  
> wrist_up  
⚡ Data_Collection_Protocol.md
```

- **Key-triggered discrete logging:** Each gesture is isolated into a specific folder, with distinct 2.5-second time windows saved as individual .txt files
- **Elimination of muscle fatigue:** Physical pause between collection effectively resets the tissue structure and prevents baseline shift
- **Strategic variance injection:** The protocol strictly standardises force gradients and arm suspension across 50 repetitions to drastically improve dataset robustness
- **Scalable and dynamically balanced:** The system allows for continuous dataset expansion beyond the initial 50 per gesture, only required same total number for each gesture

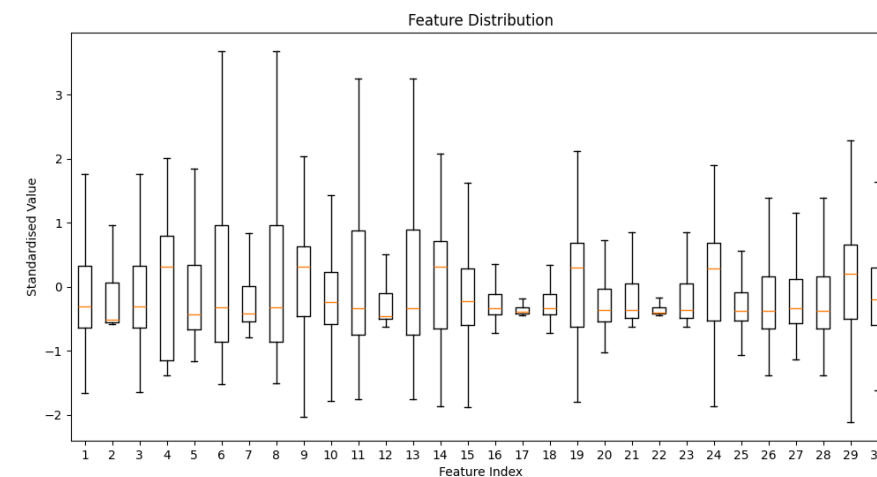


Dataset Quality & Feature



Cluster Cohesion: Data points now form highly cohesive clusters for each specific gesture, confirming that the discrete protocol successfully mitigated temporal baseline drift

Inherent Feature Overlap: While cohesive, spatial overlap remains between certain dynamic gestures (e.g., wrist up/down and fist). This objectively reflects the complex anatomical coupling of forearm muscles during contraction

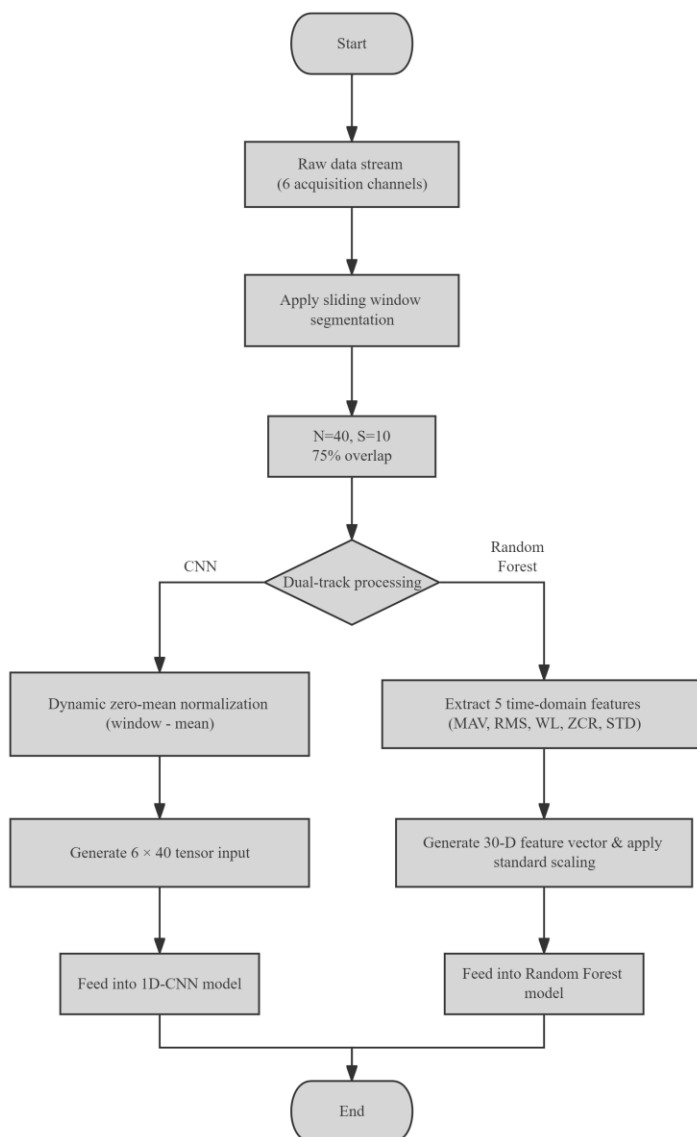


Standardised Feature Scaling: The distribution of all 30 extracted time-domain features confirms the successful application of zero-mean standard scaling

Bounded Variance: Medians align closely with the zero axis, and the interquartile ranges demonstrate bounded variance. The absence of scale-breaking outliers confirms the baseline stability of the training input



Signal Pre-processing & Classification Pipeline



Data Segmentation

- 75% overlap Ratio: The sliding window technique (N=40, S=10) yields a 75% overlap. This specific ratio is mandatory to ensure rapid, transient muscle twitches are not missed between windows
- Data Expansion: It concurrently serves to artificially expand the training dataset size from discrete recording sessions

Random Forest Track

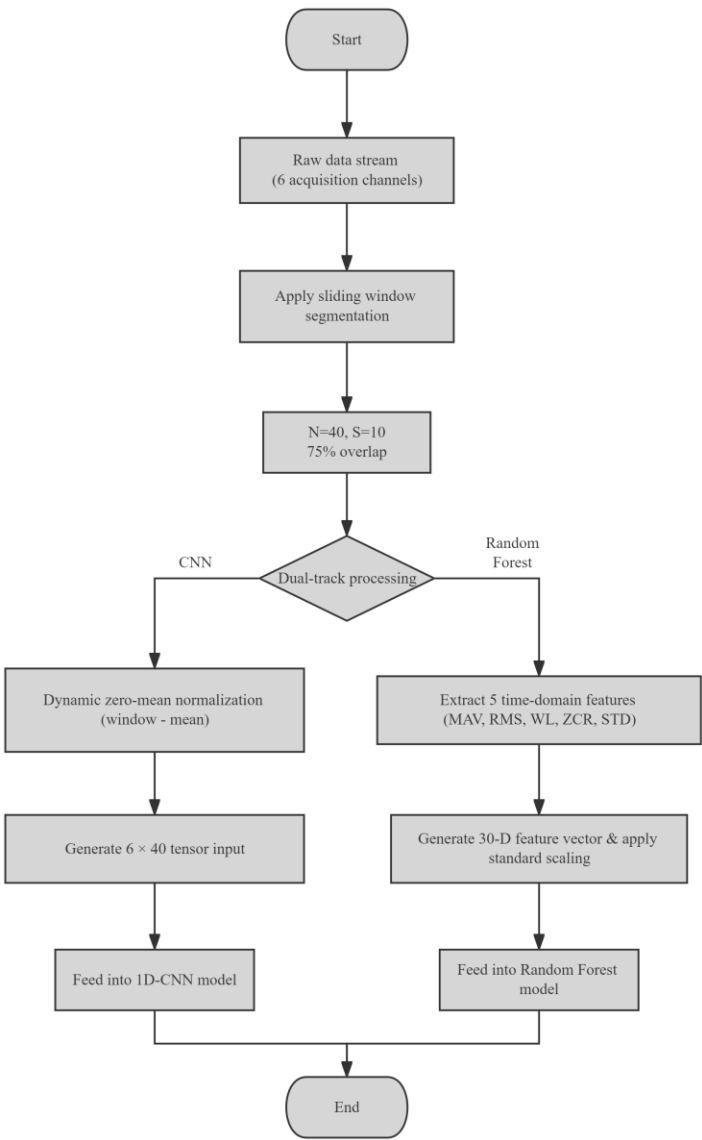
- Dimensionality Compression: Machine learning classifiers cannot effectively process raw multi-channel time-series data. The system extracts five time-domain metrics (RMS, MAV, WL, ZCR, STD) [5] to compress the raw signals into a dense 30-dimensional feature vector
- Standard Scaling: Applying standard scaling forces a zero mean and unit variance, preventing metrics with naturally large magnitudes (e.g., Waveform Length) from statistically dominating the decision trees

1D-CNN Track

- Dynamic Zero-Mean Normalization: Before feeding the 6x40 tensor into the neural network, the system subtracts the local mean of each specific window
- Isolating Waveform Topography: This operation actively strips away absolute physical baseline drift caused by sensor displacement, ensuring the CNN exclusively learns the relative shape of the muscle deformation



Offline Model Performance & Overfitting Analysis



Random Forest (Full Dataset)

True Gesture	open_hand	fist	pinch	wrist_up	wrist_down
	1919	0	0	0	0
	0	1988	0	0	0
	0	0	1905	0	0
	0	0	0	1935	0
	0	0	0	0	1941
Predicted Gesture					

1D-CNN (Test Split)

True Gesture	open_hand	fist	pinch	wrist_up	wrist_down
	274	33	49	22	6
	44	303	33	15	3
	90	31	212	44	4
	55	24	45	256	7
	12	16	14	16	330
Predicted Gesture					

Comparison Dimension	Random Forest (RF)	1D-CNN
Offline Accuracy	99.94% via overlapped / Mixed Data	70.95% via Independent Unseen Data
Data Reality	Exhibits overfitting due to dataset size limitations	Displays realistic confusion margins
Real-time inference	Fails to process live signals; gets 'stuck' between gestures and ignores user input	Handles drift effectively, but suffers from latency and relatively low real-time accuracy



Real-Time Performance & System Safety



The Accuracy-Latency Trade-off: The required 40-sample window creates an unavoidable physical buffer delay. Further parameter tuning is necessary to locate the optimal equilibrium between real-time responsiveness and classification accuracy



Temporal Flickering & Model Capacity: The 1D-CNN occasionally fluctuates during physical gesture transitions. This is not due to electrical noise, but because the current dataset volume and the lightweight baseline network capacity are insufficient to fully generalize complex boundary states



Safety-Critical Decision: Although the CNN demonstrates clear viability, its current dynamic reliability does not meet the rigorous safety standards required for human-machine interaction (HMI). Consequently, the planned physical actuation of the TetherIA bionic hand was strictly cancelled to prevent unpredictable execution and hardware damage



Summary



Low-cost Hardware Integration & FMECA

Successfully assembled the 7-DOF TetherIA tendon-driven hand without custom CAD modifications, interfaced an off-the-shelf magnetic muscle deformation armband to bypass sEMG limitations, and compiled a comprehensive FMECA report to identify and evaluate critical thermal and structural bottlenecks



Data Synchronisation Protocol & GUI

Developed a multithreaded Python GUI operating at 31 Hz to centralise data acquisition. Restructured the pipeline into a discrete, key-triggered protocol that assigns ground-truth labels directly via keystroke, completely bypassing the initial vision-based tracking to eliminate synchronisation lag and inherent recognition errors, thereby constructing a highly aligned and reliable dataset



Overfitting Identification & Drift Mitigation

Identified and exposed the "overfitting illusion" of traditional models like Random Forest under overlapping sliding windows. Successfully validated a 1D-CNN architecture integrated with dynamic zero-mean normalisation to actively strip physical baseline drift for real-time inference

Limitations & Future Work

System Tier	Targeted Future Work
Front-End (Data acquisition unit etc.)	Scale robust, key-triggered data acquisition protocols to expand dataset volume. Ensure strict data hygiene as a fundamental prerequisite for training highly generalisable models
Mid tier (Control Algorithm etc.)	Conduct extensive hyperparameter optimisation on the 1D-CNN to achieve an equilibrium between classification accuracy and processing latency. Finalise GUI command modules to transmit reliable actuation signals to end effector
Hardware tier (Bionic hand and armband etc.)	Redesign mechanical components of the TetherIA hand to resolve thermal bottlenecks identified in the FMECA. Optimise the physical fixation of the magnetic armband to ensure consistent sensor placement and reduce mechanical signal drift



Thank You for Your Attention

Open for Q&A

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References

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- [2] 'BrainCo Hand Documentation', BrainCo Hand Documentation. [Online]. Available: <https://www.brainco-hz.com/docs/revolimb-hand/>
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